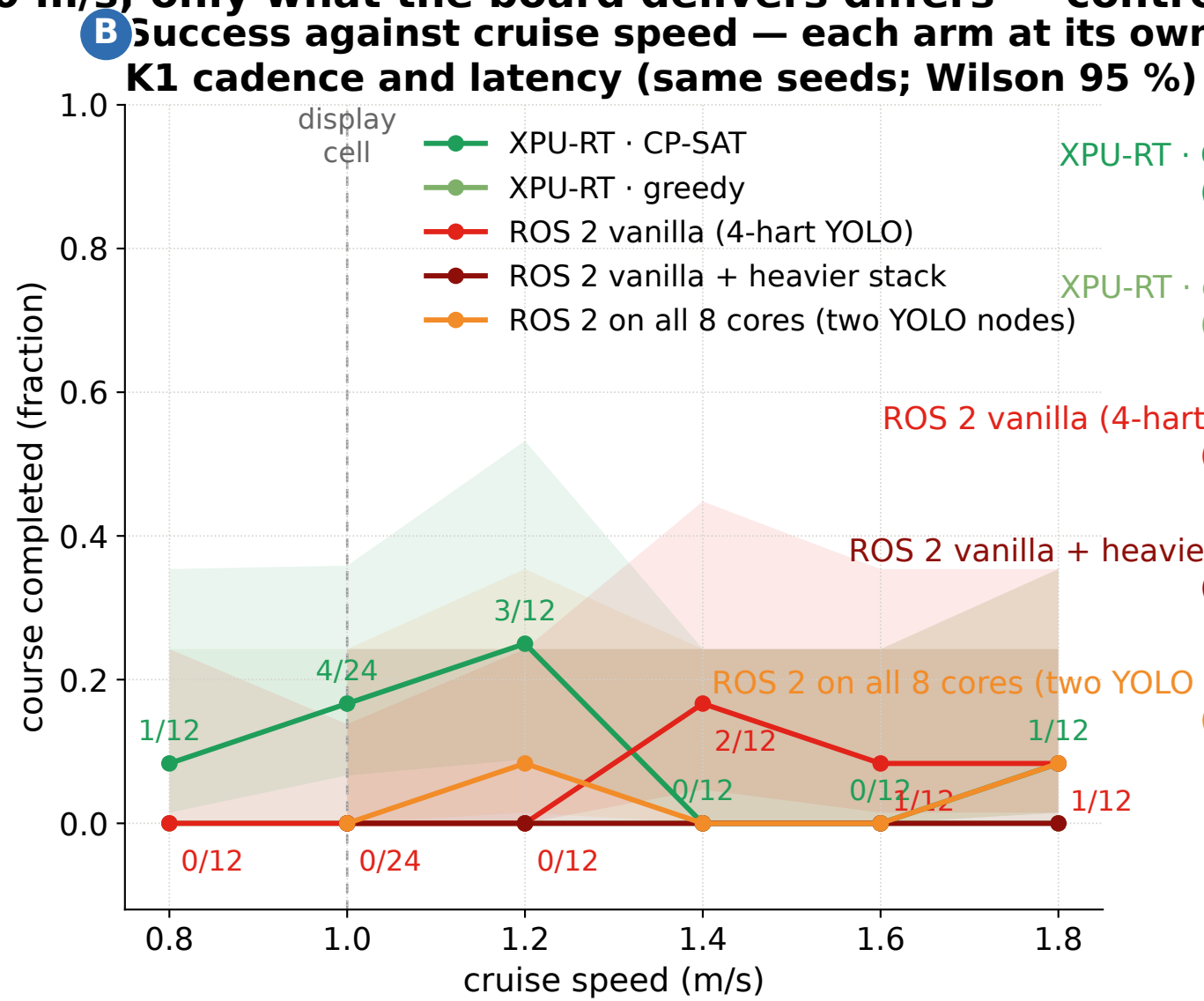
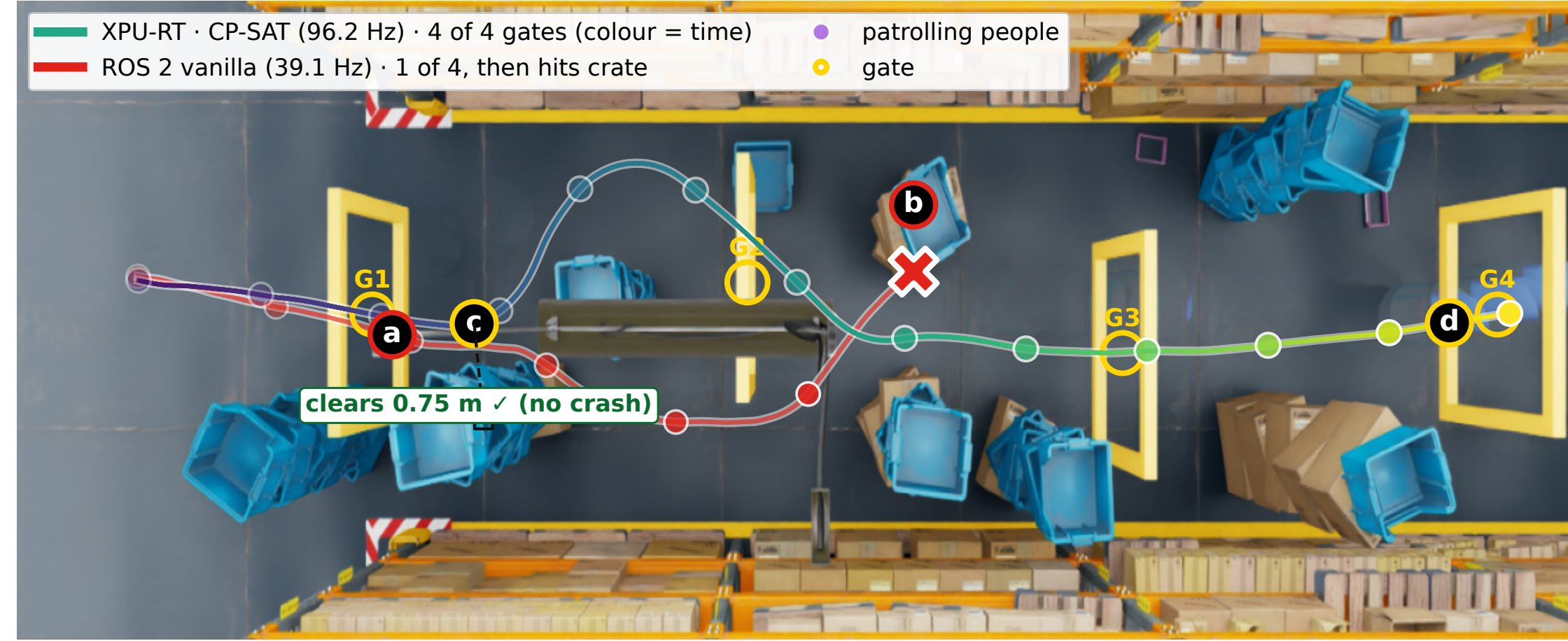
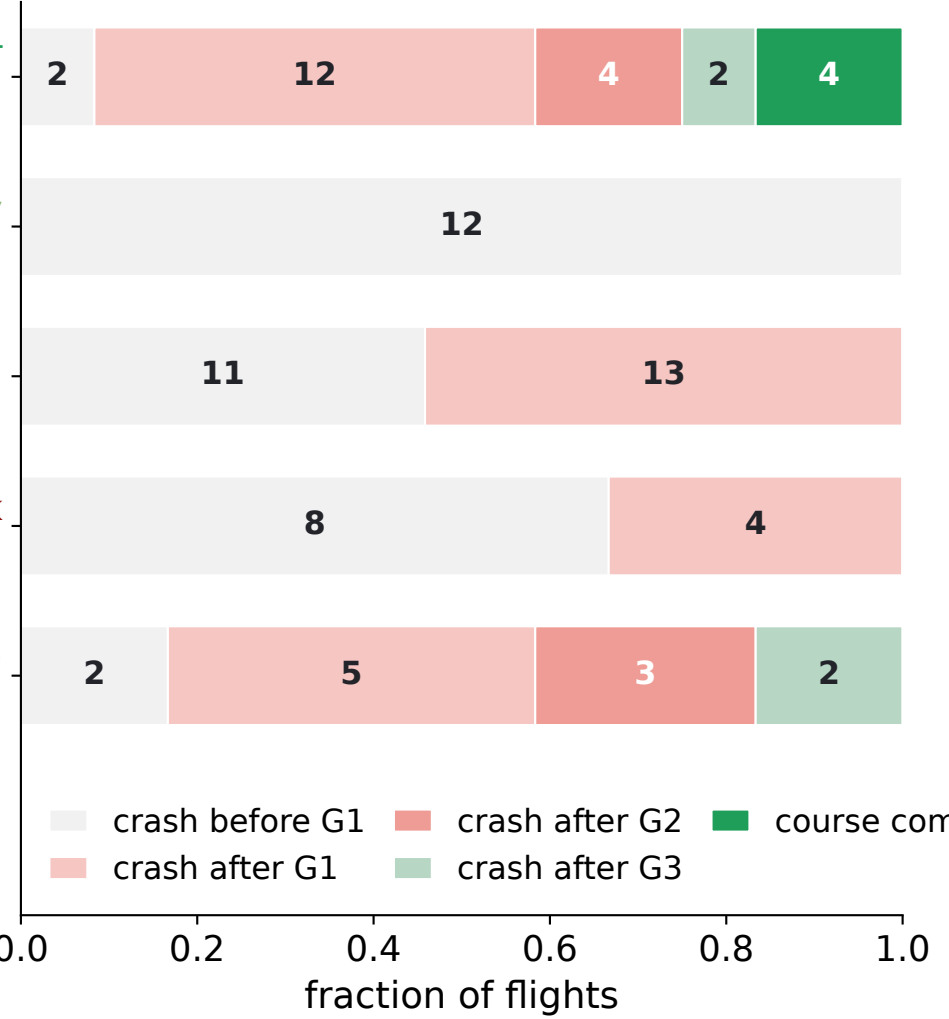


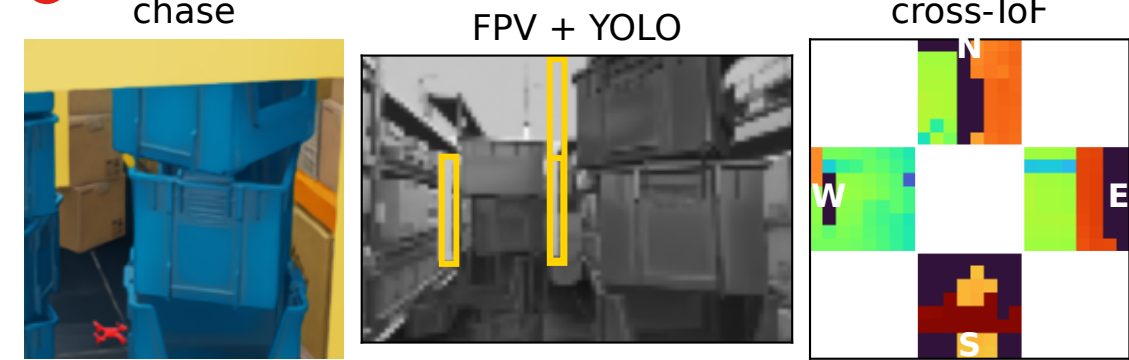
**Warehouse gate course on the K1 — same scene and controller for both runtimes at 1.0 m/s — only what the board delivers differs — control every 10.4 vs 25.5 ms, camera→control latency and goal cadence, replayed**



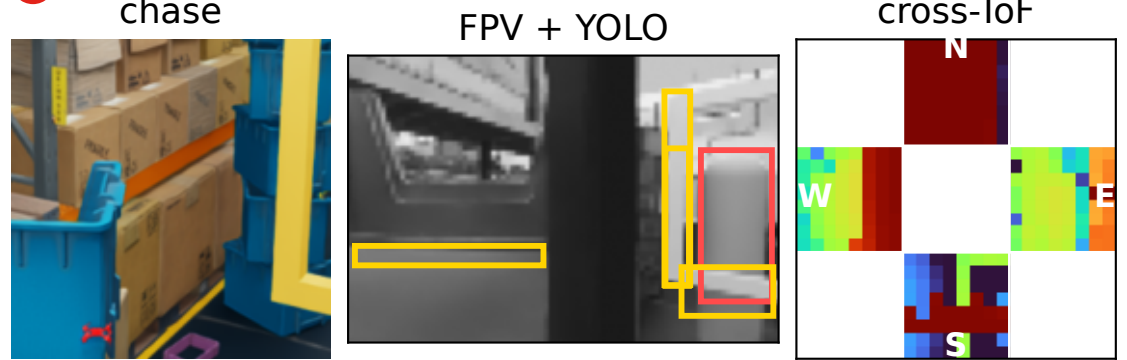
**How far each arm gets at 1.0 m/s (gates cleared before the collision)**



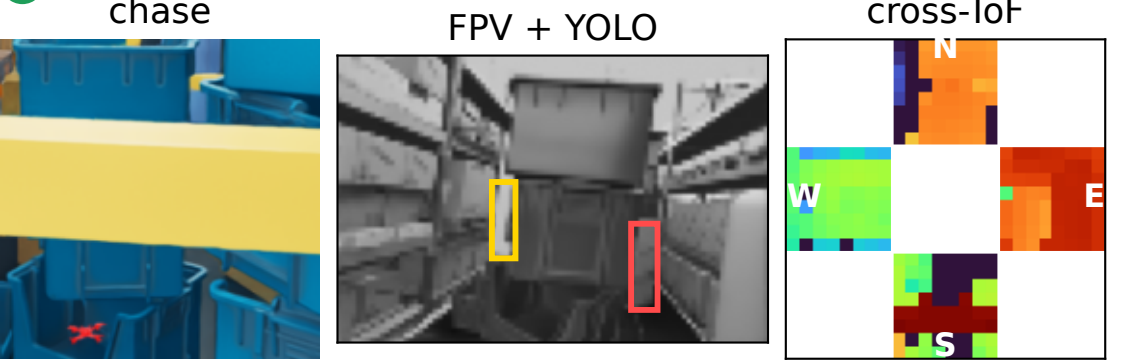
**a ROS 2 39 Hz · clears G1 · t=3.5s**



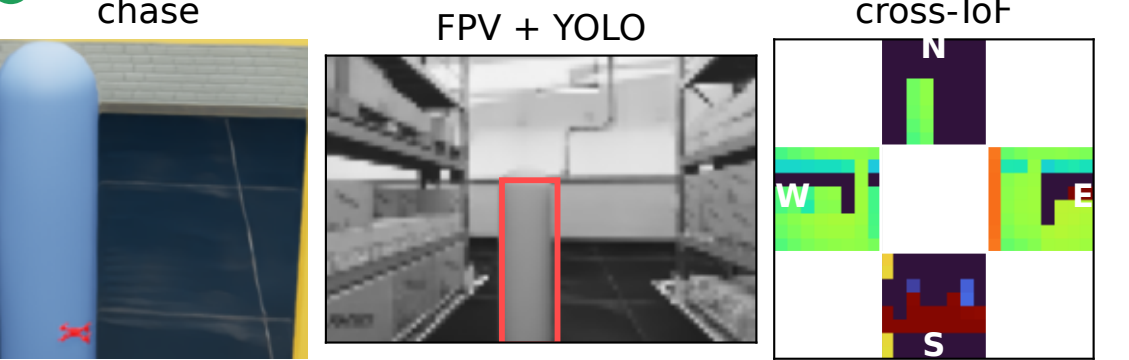
**b ROS 2 39 Hz · hits crate after G1 · t=9.5s**



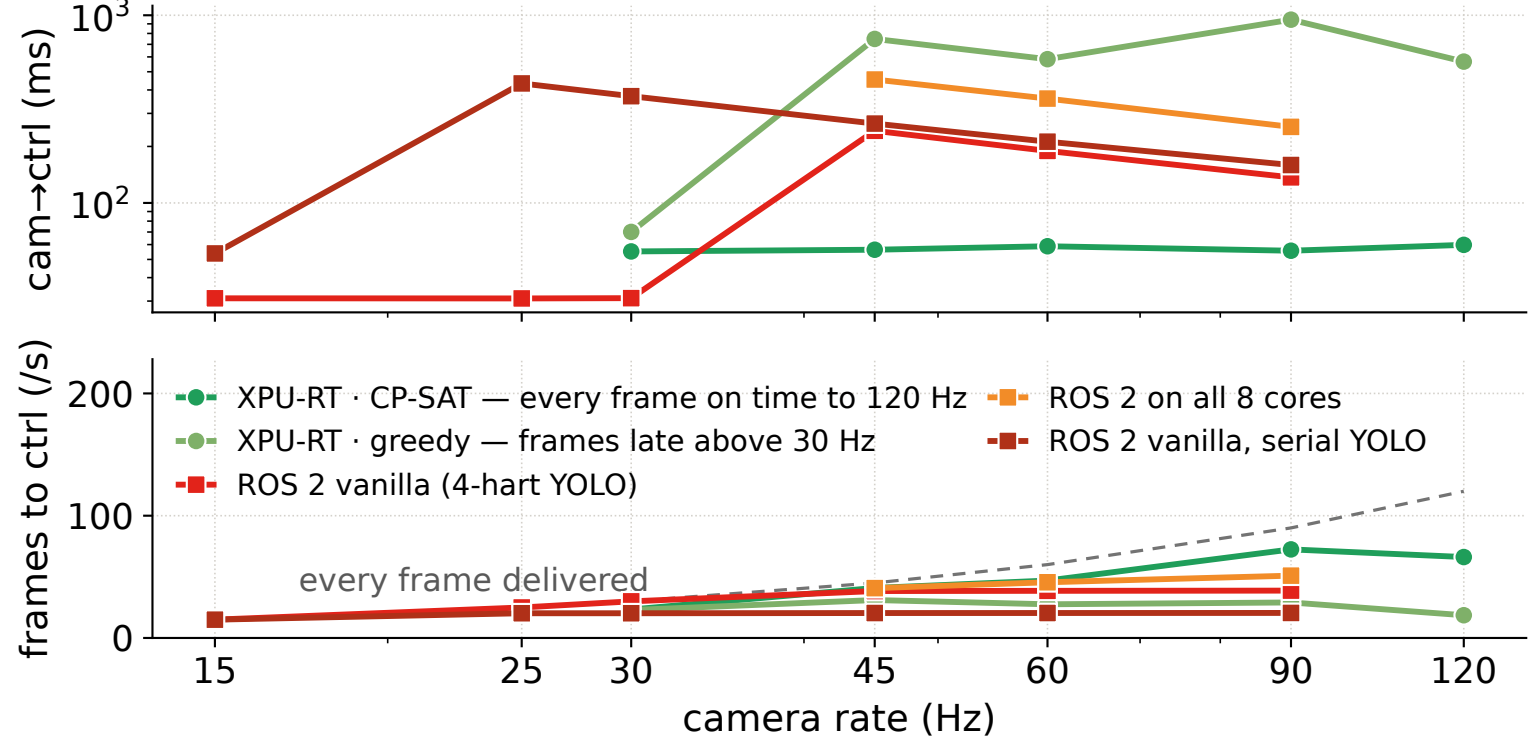
**c XPU-RT 96 Hz · clears crate 0.75 m · t=4.3s**



**d XPU-RT 96 Hz · reaches gate · t=15.3s**



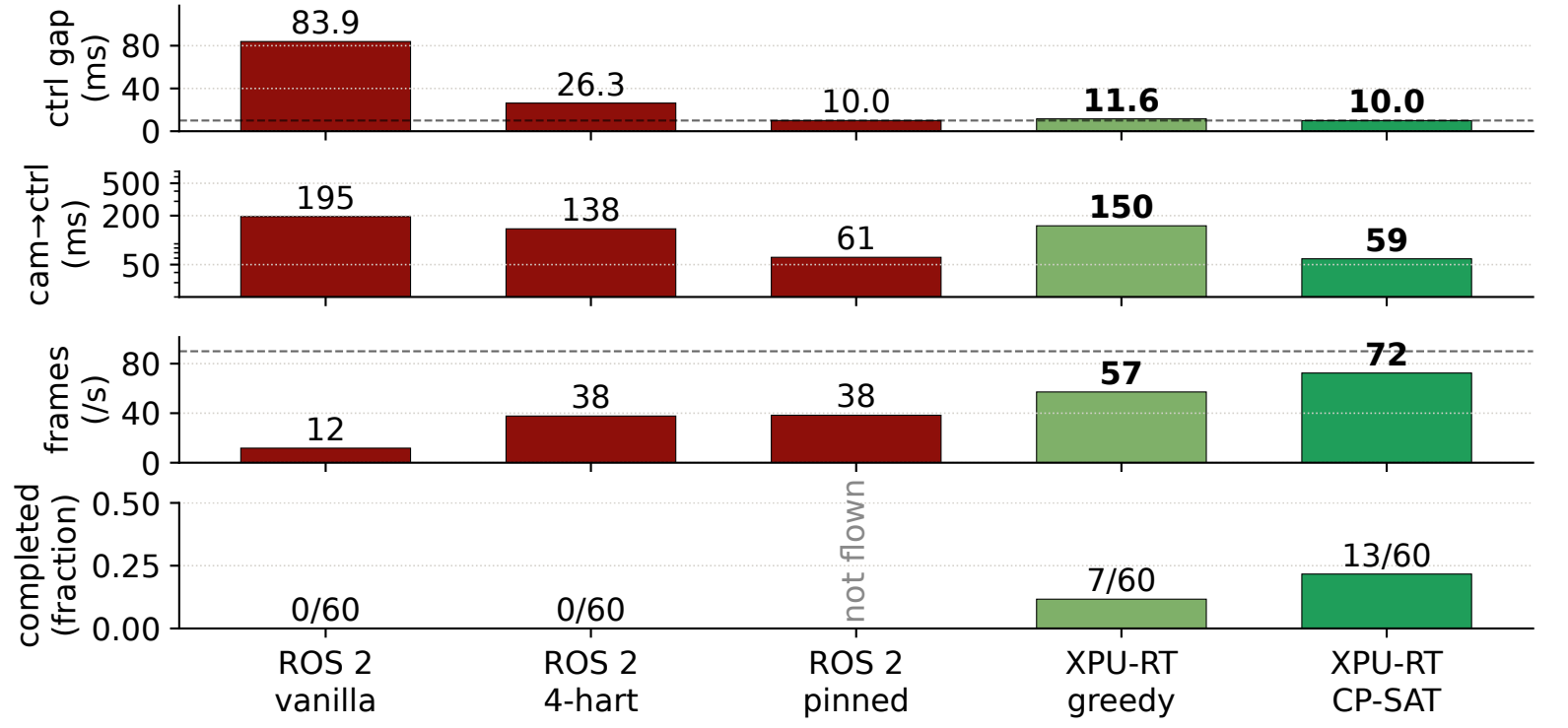
**D Camera rate on the K1: camera→control latency and frames delivered, same kernels**



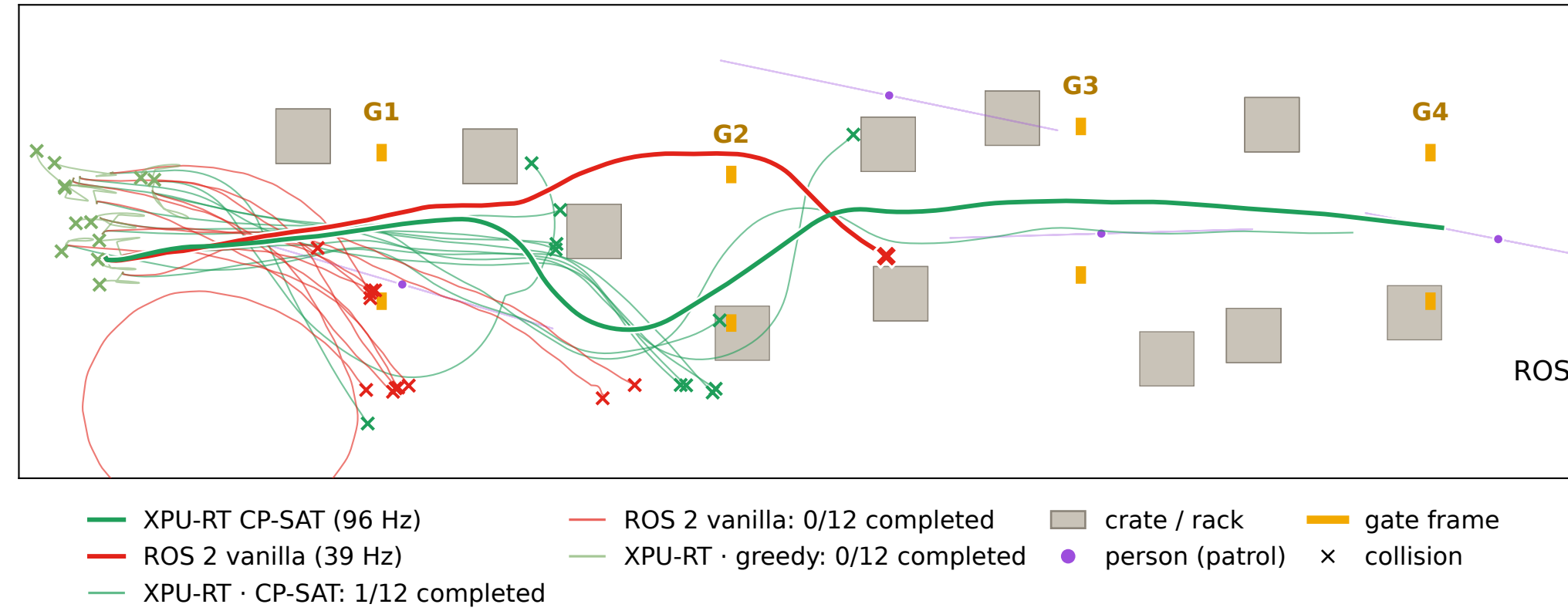
**E Where the work lands, 45 Hz camera (board traces, three runs pooled)**



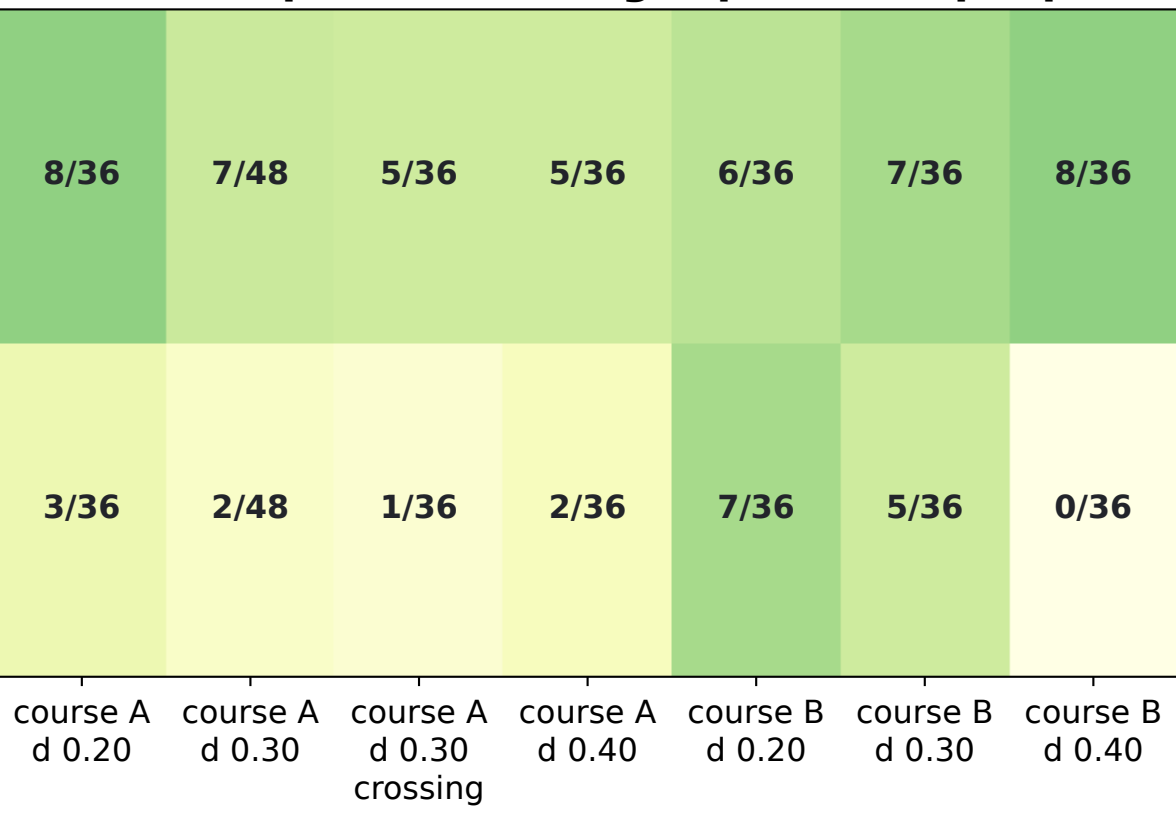
**F Added load: 90 Hz camera + ffn\_block + drone on the same 8 harts**



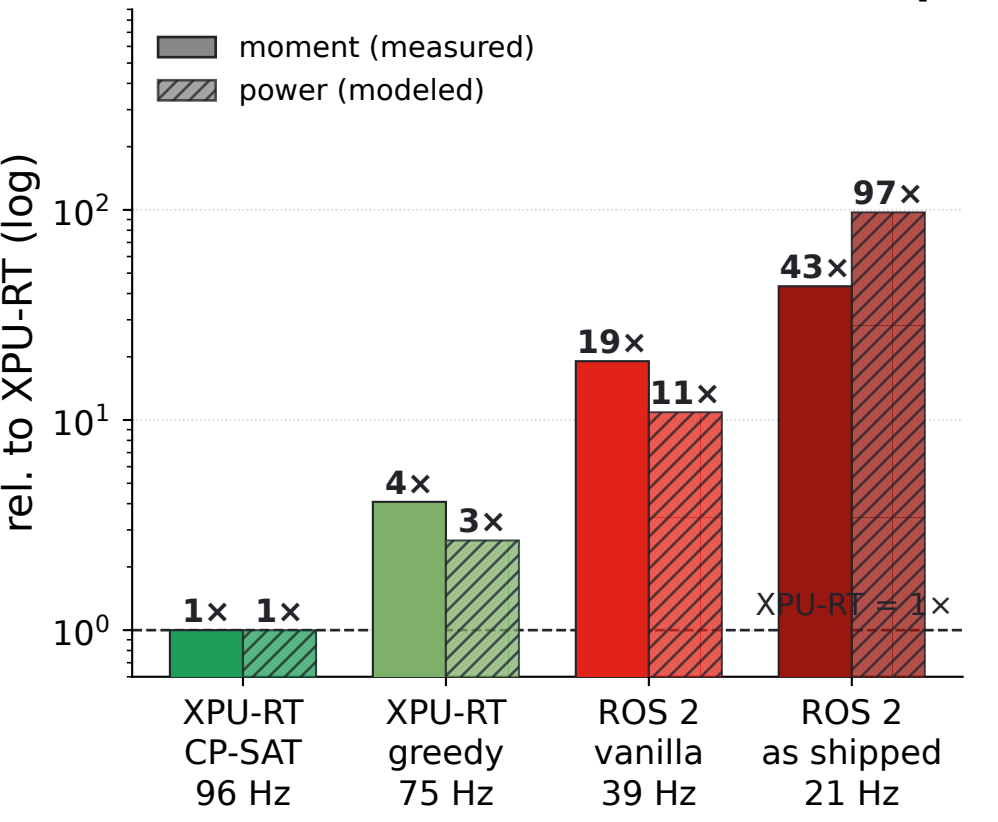
**G The display scene, every recorded run (36 flights): obstacles, people, gate frames, collisions**



**H Environments both arms flew: course completed, 1.0–1.4 m/s pooled, one flight per seed, people 2.4 m**



**I Mechanism: the baseline thrashes (commanded moment, modelled power)**



**J Onboard K1, measured, 45 Hz camera — CP-SAT: 57 ms, every frame on time, control 10.0 ms · greedy: 748 ms, 44/44 late, control 13.2 ms  
ROS 2 8-core: 447 ms, control 16.6 ms · ROS 2 vanilla: 242 ms, control 25.6 ms**

